MA 301

Test 3

Fall - 2014

Problem	points	out of
1		8
2		8
3		8
EC		2
Score		24

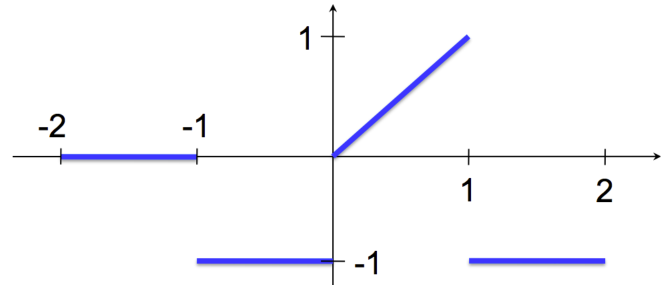
Name: ____________________

HR: ____________________

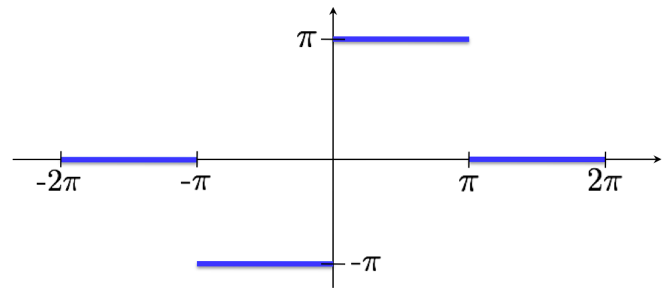
Signature: ____________________

READ BEFORE YOU START
* Show all work.
* Change your calculator mode to RADIANS.
* Sloppy solutions may be penalized.
* Use calculators ONLY for computations.
* You must write up every problem in a clear and organized way.
* If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.



Problem 2: Compute the Fourier Series for the function below. Assume the function periodically extends along the x-axis.



Problem 3: Consider the periodic function $f(x) = \begin{cases} e^{-x} & -\pi \le x \le 0 \\ e^{x} & 0 < x \le \pi \end{cases}$ and $f(x) = f(x + 2\pi)$

(a) Plot $f(x)$ on the interval $[-\pi, 3\pi]$ *(yes, I meant* 3π*)*

(b) Find the Fourier Series of $f(x)$

$\left[\textit{Hint: An anti-derivative of } e^x \cos(nx) \textit{ is } \dfrac{e^x}{1+n^2}\left(\cos(nx) + n\sin(nx)\right)\right]$

Extra Credit: Show that $\dfrac{e^x}{1+n^2}(\cos(nx) + n\sin(nx))$ is an anti-derivative of $e^x \cos(nx)$

Note: Do not just take the derivative of $\dfrac{e^x}{1+n^2}(\cos(nx) + n\sin(nx))$